Here's a detailed **table of AI models**, grouped by category, with explanations and examples so you get a **clear and complete view** of the AI landscape.

AI Model Type	Description	Examples	Real-World Use Cases
1. Machine Learning (ML) Models	Models that learn patterns from data and make predictions without explicit programming.	**Linear Regression** – predicts continuous values.**Logistic Regression** – predicts categories.**Decision Trees** – splits data into branches for decision-making.**Random Forest** – combines multiple trees for better accuracy.**K-Nearest Neighbors (KNN)** – classifies based on closest data points.	Predicting house prices (Linear Regression).Spam detection (Logistic Regression).Loan approval (Decision Trees).Customer churn prediction (Random Forest).
2. Deep Learning (DL) Models	A subset of ML using **artificial neural networks** with multiple layers for complex patterns.	**Feedforward Neural Networks (FNN)** – basic neural network.**Convolutional Neural Networks (CNN)** – image processing.**Recurrent Neural Networks (RNN)** – sequence data.**Long Short-Term Memory (LSTM)** – improved RNN for long sequences.**Transformer Models** – attention mechanism for text and vision.	Image recognition (CNN).Speech recognition (RNN, LSTM).Machine translation (Transformers).Stock trend forecasting (LSTM).
3. Natural Language Processing (NLP) Models	AI models that understand, interpret, and generate human language.	**BERT** – bidirectional transformer for understanding text.**GPT** – generative transformer for text generation.**T5** – text-to-text tasks.**Word2Vec / GloVe** – word embeddings.	Chatbots (GPT).Sentiment analysis (BERT).Text summarization (T5).Search engines (Word2Vec).
4. Generative Models	Models that create new data similar to training data.	**Generative Adversarial Networks (GANs)** – two networks compete to generate realistic data.**Variational Autoencoders (VAEs)** – learn data distribution to generate samples.**Diffusion Models** – iterative noise removal to create high-quality images.	Deepfake videos (GANs).AI art creation (Diffusion Models like DALL·E, Midjourney).Data augmentation (VAEs).

5. Reinforcement Learning (RL) Models	Models that learn by interacting with an environment and receiving rewards or penalties.	**Q-Learning** – table-based RL.**Deep Q-Networks (DQN)** – deep learning with Q-learning.**Policy Gradient Methods** – directly learn policies.**Actor-Critic Methods** – combine value and policy learning.	Self-driving cars (DQN).Game-playing AI like AlphaGo (Policy Gradient).Robotics navigation (Actor-Critic).
6. Computer Vision Models	AI models specialized for understanding and interpreting visual information.	**YOLO (You Only Look Once)** – real-time object detection.**ResNet** – deep CNN for image classification.**Mask R-CNN** – object detection + segmentation.**EfficientNet** – optimized image recognition.	Face recognition (ResNet).Security camera monitoring (YOLO).Medical image analysis (Mask R-CNN).
7. Hybrid & Multi-Modal Models	Models that process multiple types of input (text, image, audio) together.	**CLIP** – connects images with text descriptions.**BLIP** – vision-language model.**GPT-4 with Vision** – interprets text and images.**Flamingo** – handles multiple modalities in conversation.	Image captioning (CLIP).Visual question answering (BLIP).Multi-modal chatbots (GPT-4 Vision).
8. Specialized Domain Models	AI trained for specific fields.	**AlphaFold** – predicts protein structures.**Med-PaLM** – medical language understanding.**Codex** – AI code generation.	Drug discovery (AlphaFold).Medical diagnosis assistance (Med-PaLM).Code writing (Codex).

If you like, I can also prepare a **second table that shows each AI model type with step-by-step working and a small Python example** so you not only know *what* they are but *how* they work internally. That way, you'll have a **practical + theoretical AI reference sheet**.